
Correct Scores, Wrong Scientific Conclusions: Auditing Learned Surrogates of Chaotic Dynamics

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Abstract

1 Learned forecasters increasingly stand in for scientific models that are themselves
2 imperfect, and a forecast score alone cannot say whether the surrogate succeeds
3 for the right reasons. We test five conditions that connect a score to a scientific
4 claim (held-out generalization, complete failure accounting, matched disclosure,
5 structural identifiability, and cross-system transfer) on k -pendulum ($k \in \{1, 2, 3\}$)
6 and Lorenz-63 forecasts from language models, a time-series foundation model,
7 learned dynamics, sparse identification, and classical integrators. Every test over-
8 turns a conclusion the raw comparison invites: the in-sample/out-of-sample gap
9 is fourfold, two exact symmetries make the apparent parameter recovery impos-
10 sible, and a second system reverses the model ordering. A correct leaderboard
11 can still support incorrect scientific conclusions. We release the prompts, outputs,
12 checkpoints, seeds, and compute records needed to repeat the audit.

13 1 Introduction

14 Scientific claims require more than low forecast error. When a learned forecaster stands in for an
15 imperfect scientific model, a result may depend on training trajectories, dropped failures, unmatched
16 inputs, an unidentifiable target, or one favorable system: failures of the evaluation rather than of the
17 surrogate. We study these failure modes directly. The task is to predict a chaotic system’s state s_T
18 from its state s_0 , on two systems chosen to disagree: the k -link pendulum, chaotic for $k \geq 2$, with
19 high-precision ground truth and control over what physical metadata a model is told, and Lorenz-63,
20 which differs in observation model and hypothesis class.

21 Our contribution is a claim audit for dynamical-system forecasts: held-out generalization, complete
22 failure accounting, matched disclosure on identical trajectories, identifiability checked before scoring
23 recovery, varied training budgets, and transfer to a second system. Table 1 states the conclusion
24 before and after each test; all five alter or reject a conclusion the unaudited comparison supports.
25 These are conclusions the field actually draws: zero-shot forecasters are routinely scored without
26 failure accounting or matched controls [Gruver et al., 2023, Zhang and Gilpin, 2025], and our setting
27 mirrors that practice. Three of the checks are the familiar concerns of imperfect models applied to the
28 evaluation itself: misspecification (a Hamiltonian prior on damped dynamics), identifiability (exact
29 parameter degeneracies), and transfer (a hypothesis class containing one system and not the other).

30 Appendix D situates the audited model families and the identifiability and selective-prediction
31 background.

Table 1: Conclusions before and after the claim audit.

Test	Unaudited conclusion	Audited conclusion
Generalization	Learned dynamics forecast accurately.	Error rises about fourfold; simple baselines perform better.
Failures	Answered forecasts represent model performance.	Incomplete coverage lowers reliability-adjusted performance.
Disclosure	Withholding constants improves forecasts.	Withholding constants increases pooled error by 0.069 rad.
Identifiability	Absolute constants can be recovered from angles.	Two exact symmetries make absolute recovery impossible.
Transfer	The pendulum order ranks model families.	Lorenz-63 produces a different order.

2 Benchmark Design

2.1 Task and systems

The pendulum state is $\mathbf{s}_0 = (\theta_1, \dots, \theta_k, \omega_1, \dots, \omega_k) \in \mathbb{R}^{2k}$; hiding its constants does not change the trajectory, which permits the matched test in §4.2. Ground-truth trajectories integrate the full nonlinear Lagrangian equations of motion with fixed-step RK4 at $\Delta t = 10^{-3}$ s, recording states every 0.01 s plus a 10 s pre- $t=0$ context window (used only by the time-series models). Initial conditions are sampled $\theta_i \sim \mathcal{U}(-3, 3)$ rad, $\omega_i \sim \mathcal{U}(-1, 1)$ rad/s with a fixed seed; because the labels come from RK4, the coarser-step RK4 baseline is a reference by construction, not a competitor.

2.2 Regimes and grid

Three regimes test generalization to non-standard constants (Table 8); in `changed_hidden` the constants (g, L, m , damping) are not disclosed to the model. An evaluation cell is a tuple (model, k , regime, horizon), with horizons $\{0.01, 1, 10, 60\}$ s. The full LLM grid also varies input modality and prompting; every comparison here uses the coordinate, direct setting, and the other axes appear only in the released records.

Out-of-sample protocol. Learned models are evaluated off their training data: a shared held-out set (a separate seed) of 180 initial conditions (60 per regime, $k \in \{1, 2, 3\}$) is scored identically for every family. The cross-family comparison uses horizons $\{1, 10\}$ s, where chaos has begun but error has not saturated ($180 \times 2 = 360$ cells per model); in-sample error appears only once, to quantify the size of the generalization gap (§4.1).

2.3 Inputs and grouping

Each family sees different information, and we treat that as part of the experimental record (Table 3, Appendix B). The integrators get the exact equations and constants; the learned models get target-system training plus the constants (normal-regime priors in hidden cells); the LLMs get a text description with the constants except in hidden cells; Chronos gets a 10 s pre- $t=0$ history window and no constants at all. A further asymmetry runs the other way: the LLMs are large proprietary models while the learned models are small MLPs (Appendix B), so larger physics-informed networks are outside our scope. We therefore separate known-physics methods (RK4, Euler, symplectic), which serve as numerical references, from the black-box methods being compared.

2.4 Metrics

The primary error is the mean absolute angle error (radians), wrapped to $[-\pi, \pi]$ and averaged over links and trajectories.

Reliability-adjusted error. A cell is answered only if the model returns a parseable, finite prediction. Because scoring answered cells alone would reward abstention on hard cells, unanswered cells receive the random-guess baseline $\pi/2$ (the expected wrapped-angle error of a uniform guess); answering is part of the score.

67 **Verification record.** Each cell retains its identifiers, prompt, raw response, parser outcome, pre-
68 diction, token counts, and score. Local methods are reproducible from the released checkpoints and
69 seeds; hosted deployments can change or be withdrawn, so for them the released per-cell responses
70 are the record, keeping the API evaluation auditable after the deployments it measured are gone.

71 **Disclosure gap.** We define $\Delta_{\text{disc}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$; positive values indicate a penalty
72 for withholding constants. Because regime contrasts confound disclosure with trajectory difficulty,
73 §4.2 also toggles disclosure on identical trajectories.

74 3 Methods

75 **LLMs.** We evaluate `kimi-k2.6`, `grok-4-1-fast-reasoning`, and `deepseek-v4-pro` through
76 an OpenAI-compatible API (Azure AI Foundry deployments; evaluated June 2026), with a structured
77 prompt and JSON answer schema (Appendix A); each cell is one temperature-0 call with uniform
78 token and timeout budgets, no best-of- N .

79 **Time-series foundation models.** We run Chronos [Ansari et al., 2024] locally in its standard
80 history-conditioned form: the 10 s pre- $t=0$ context window of all $2k$ state channels forecasts to the
81 target horizon. The comparison uses `chronos-2` in univariate and multivariate modes; the paired
82 mode comparison uses 20 trajectories per (k, regime) cell.

83 **Learned dynamics.** Neural ODE, HNN, and LNN share a 256-wide, 3-layer tanh MLP and
84 integrate at inference with `scipy`, comparing derivative matching, rollout, and mixed- k rollout
85 objectives (Appendix B). The primary models train on nine trajectories, one per (k, regime) cell;
86 a nested budget study trains fresh checkpoints at $\{1, 5, 20\}$ trajectories per eligible cell with three
87 initializations (HNN: undamped cells only, since damping violates Hamiltonian conservation).

88 **Integrators.** Euler, RK4, and Störmer–Verlet (symplectic) at $\Delta t=0.01$ s using the exact equations
89 of motion; their error is from discretization alone.

90 **Baselines.** Four reference baselines use the same grid: persistence ($\theta_T=\theta_0$), constant velocity
91 ($\theta_T=\theta_0+\omega_0 T$), a small-angle linearized model, and a nearest-neighbor analog forecast built from the
92 same data as the learned models. On the balanced two-seed subset we also fit SINDy (degree-two
93 library, STLSQ) and NVAR (two delays, degree-two features, ridge); divergent rollouts are retained
94 as failures at $\pi/2$.

95 **Lorenz replication.** A secondary Lorenz–63 test compares numerical references, persistence,
96 SINDy, NVAR, a Neural ODE trained on 20 seed-42 trajectories, and zero-shot Chronos-2 on five
97 five-trajectory held-out seeds, reporting training-normalized coordinate MAE at $\{0.1, 0.5, 1.0\}$ s
98 (configuration in Appendix C). HNN and LNN are omitted because Lorenz–63 is dissipative and
99 non-Hamiltonian.

100 4 Results

101 4.1 Counting failed responses

102 Every model is evaluated on the shared held-out grid of 360 cells (180 trajectories, horizons $\{1, 10\}$ s);
103 the Answered column in Table 2 gives the fraction with valid predictions. `grok` answers 82% and
104 `l1nn` 94%; their ranks fall when missing cells receive $\pi/2$; a mean over answered cells alone would
105 flatter both. Intervals are 95% bootstrap over trajectories; LLM intervals exclude per-call sampling
106 variation.

107 Table 2 ranks the models under the access conditions of §2.3; the ranking is conditional on those
108 inputs, this training budget, and this system (§4.4 tests transfer). The known-equation integrators
109 sit at the top as references, and every learned dynamics model lands closer to the $\pi/2$ baseline than
110 to the integrators. The lowest-error black-box entry is the LLM `kimi` (0.538 rad, 99% answered),
111 and its marginal interval excludes the nearest baseline’s. That is a statement about one recorded
112 June 2026 deployment, not about model families; the Lorenz test rejects the family reading.

Table 2: Out-of-sample cross-family comparison (360 cells, horizons $\{1, 10\}$ s): reliability-adjusted mean angle error (rad), unanswered cells at $\pi/2$, 95% bootstrap CI. RK4 generates the data (reference). †: configurations first evaluated in this work; ‡: simple baselines.

#	Model	Err	95% CI	Answered	Family
–	rk4 (reference)	0.059	[0.032, 0.089]	100%	numerical
1	symplectic	0.205	[0.163, 0.251]	100%	numerical
2	euler	0.405	[0.344, 0.470]	100%	numerical
3	kimi-k2.6	0.538	[0.475, 0.602]	99%	LLM
4	chronos2-multi†	0.684	[0.617, 0.752]	100%	time-series
5	chronos2-uni†	0.697	[0.631, 0.766]	100%	time-series
6	nearest-neighbor‡	0.725	[0.657, 0.797]	100%	baseline
7	linearized‡	0.855	[0.774, 0.939]	100%	baseline
8	grok-4-1-fast-reasoning	0.881	[0.805, 0.957]	82%	LLM
9	deepseek-v4-pro	0.952	[0.881, 1.027]	100%	LLM
10	hnn-rollout	0.972	[0.898, 1.047]	100%	learned
11	persistence‡	1.004	[0.934, 1.078]	100%	baseline
12	neural-ode-rollout-mixed†	1.036	[0.967, 1.104]	100%	learned
13	hnn-rollout-mixed†	1.066	[0.990, 1.143]	100%	learned
14	neural-ode	1.110	[1.039, 1.181]	100%	learned
15	neural-ode-rollout	1.121	[1.054, 1.190]	100%	learned
16	hnn	1.138	[1.059, 1.218]	100%	learned
17	constant-velocity‡	1.319	[1.241, 1.396]	100%	baseline
18	lnn† (full-scale)	1.356	[1.282, 1.430]	94%	learned

After equalizing the hidden-regime inputs, nearest-neighbor lookup (0.725) and the linearized pendulum (0.855) beat every learned model, and persistence (1.004) beats all but HNN-rollout (0.972), though those two intervals overlap. The stable learned models score about 0.26 rad on their training trajectories and 0.97–1.14 rad held-out, so in-sample evaluation understates error about fourfold.

SINDy (0.883 rad) and NVAR (1.248) appear only on the balanced two-seed subset (Table 4); SINDy lands between Chronos and the learned models, NVAR near constant velocity.

4.2 Matched disclosure test

On the shared held-out set the hidden regime shows lower error than the disclosed (-0.390 rad, $p_{\text{Holm}} < 10^{-3}$, unpaired) and normal (-0.524) regimes. This looks favorable to withholding constants, but the regimes use different trajectories, and the hidden ones are slower and less divergent. When we instead toggle disclosure on identical trajectories (Table 7), the pooled effect reverses: hiding the constants increases error ($\Delta_{\text{disc}} = +0.069$ rad, $p_{\text{Holm}} = 0.034$), with disclosure helping strongly at $k=1$ and only a small reverse effect at $k=3$. The unadjusted hidden-regime advantage is thus a difficulty confound.

The disclosure test also has a structural limit. The angular equations are unchanged by $(m, d) \mapsto (am, ad)$ and by $(L, g, d) \mapsto (bL, bg, b^2d)$, so for $k=1$ only g/L and $d/(mL^2)$ are identifiable from an angle trajectory; absolute g , L , m , and d cannot be scored as independently recoverable. Asked to name the constants from a single state, the LLMs mostly returned Earth-standard defaults, which is evidence of a prior, not of recovery; a released numerical check confirms both invariances hold to below 10^{-13} .

4.3 Replication across seeds and training budgets

We reran the local pendulum subset on an independent held-out seed (five trajectories per cell versus 20; Table 4). The selected ordering is unchanged: numerical references lowest, Chronos-2 before the learned models, HNN-rollout best among them. Two seeds do not estimate seed-level uncertainty. The LNN’s low primary-seed coverage was a hardware artifact: its per-step Hessian exhausted the GPU on long rollouts. A CPU rerun recovers the dropped cells ($66\% \rightarrow 94\%$; both seeds use it); the remaining failures are genuine divergences, scored at $\pi/2$, and the two-seed mean stays 1.366 rad.

Training-budget check. Across three initializations and both held-out seeds, Neural ODE error falls from 1.090 at one trajectory per cell to 0.861 at twenty, and normal-regime HNN error from

1.269 to 0.981 (Table 9); the rows cover different regimes and are not an architecture comparison. More data helps the Neural ODE monotonically, the HNN gains plateau by 15 trajectories, and neither closes the gap to known physics.

4.4 Cross-system test: Lorenz-63

We test whether the pendulum conclusions transfer to Lorenz-63, a chaotic system with a polynomial vector field. They do not. The identifiability verdict flips: on the pendulum the physical constants are not recoverable from an angle trajectory (§4.2), while on Lorenz, SINDy recovers the governing equations themselves, not merely their forecasts. The refit selects exactly the true sparsity pattern and returns $(\hat{\sigma}, \hat{\rho}, \hat{\beta}) = (10, 28, 8/3)$ to coefficient error 2×10^{-13} (Appendix C); forecast error stays below 2×10^{-5} in every seed. The difference is the setting, not the audit: Lorenz-63 lies inside SINDy’s hypothesis class and is fully observed; the pendulum’s absolute constants are unidentifiable from angles under any method. Identifiability is a property of the observation model and the hypothesis class, and the check makes it explicit before recovery is scored.

The model ordering also changes. Trained on one seed and evaluated on five held-out seeds (25 trajectories), a Lorenz Neural ODE reaches normalized MAE 0.014 ± 0.008 , 0.079 ± 0.030 , and 0.277 ± 0.063 at 0.1, 0.5, and 1.0 s (mean $\pm$ SE over seeds), below both Chronos-2 modes at every horizon (multivariate 0.101 ± 0.018 , 0.691 ± 0.129 , 0.771 ± 0.097 ; univariate similar); NVAR drifts to 0.429 by 1.0 s and persistence trails (0.900; Table 4). This reverses the pendulum, where trained models lost to trivial baselines: here the system-trained model beats the zero-shot foundation model. The families again see different inputs, so this is not an architecture race; what the cross-system result rejects is the pendulum ranking as a general ordering of model families.

The reversal does not rest on a favorable seed: the Neural ODE beats both Chronos-2 modes in every one of the 15 seed-by-horizon comparisons, and the identifiability half is not a sampling claim at all, since the pendulum symmetries are exact and Lorenz-63 lies in SINDy’s library by construction. The test excludes the API LLMs, HNN, and LNN (a Hamiltonian surrogate is misspecified for the dissipative Lorenz flow). Those omissions limit what it can establish, not what it can refute: one system on which both verdicts invert is enough to withdraw the generalization, and the Lorenz ordering is not offered as the correct one.

4.5 Diagnostics

Appendix E reports five further checks; two matter most here. First, the physical-fidelity check localizes a misspecification: on the undamped regime the lowest learned-model energy drift (0.78, E^* -normalized) far exceeds the symplectic integrator’s (0.066), and a symplectic HNN rollout leaves drift nearly unchanged, so the fault is the learned Hamiltonian, not the integrator. Second, every black-box family saturates near the chaotic noise floor by 1 s, hence the $\{1, 10\}$ s comparison window. The rest: error rises monotonically with k ; the hidden regime is easier for most models (the §4.2 confound); rollout loss is not uniformly better than derivative matching; and Chronos-2’s two modes are indistinguishable on 720 paired samples.

5 Discussion and Conclusion

Performance depends on the system, observations, hypothesis class, data, and access, and a scalar leaderboard does not encode those conditions: each of the five checks changes a claim the raw leaderboard supports. Forecasting benchmarks should report the checks, training budgets, and information access as part of the result. We chose systems with cheap ground truth so each check could be shown to change a conclusion; the checks matter most where real observations meet an imperfect simulator, verification is expensive, and misspecification cannot be ruled out by construction. All prompts, outputs, checkpoints, seeds, and compute records accompany the paper.

Limitations. The LLM grid uses one held-out seed, the local replication a smaller second, the budget study three initializations, and the Lorenz test five trajectories per seed without the API LLMs, HNN, or LNN. SINDy’s recovery requires the library to contain the true degree, predictions are endpoint states, and pendulum-like data may appear in the foundation models’ corpora.

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A Prompt templates

The prompts used for every LLM evaluation cell are reproduced below verbatim.

Table 3: Inputs available to each family. State is at $t=0$; history is pre- $t=0$. Chronos is history-conditioned. *In changed_hidden cells the learned models receive normal-regime prior constants instead of the true ones.

Family	State	History	Constants	Equations
Numerical (RK4/Euler/sympl.)	✓	—	✓ (all regimes)	✓ (exact)
Learned (NODE/HNN/LNN)	✓	—	✓ base; withheld*	— (learned)
Time-series (Chronos)	—	✓ (10 s context)	—	—
LLMs (kimi/grok/deepseek)	✓	—	except hidden	—

234 System prompt (direct / no-CoT).

235 You are a careful physicist predicting the future state of a k -pendulum (a chain of
236 k point masses on massless rods, hinged in a planar chain, swinging under gravity
237 with viscous damping). Respond with ONLY a single JSON object on one line, no prose,
238 no code fences. Schema:
239 {"theta": [...k floats...], "omega": [...k floats...]}
240 theta_i is in radians (link i measured from straight-down,
241 positive counter-clockwise)
242 and omega_i is angular velocity in rad/s.

243 **System prompt (chain-of-thought).** Same preamble, then: “Reason step-by-step about the dynamics,
244 then output your final answer. Wrap the final answer in `<answer>...</answer>` tags containing ONLY a
245 single JSON object: {"theta": [...k floats...], "omega": [...k floats...]}.”

246 User prompt (coordinate modality).

247 Number of links $k = \{k\}$
248 Physical constants:
249 g (gravity, m/s^2): $\{g\}$
250 L (rod lengths, m): $\{L\}$
251 m (point masses, kg): $\{m\}$
252 damping (viscous coefficient on omega): $\{\text{damping}\}$
253 Initial state at $t=0$:
254 theta_0 (rad): $\{\text{theta}_0\}$
255 omega_0 (rad/s): $\{\text{omega}_0\}$
256 Predict the state at $t = \{T\}$ seconds.
257 Return theta and omega as length- $\{k\}$ JSON arrays.

258 **Hidden-constants regime.** The constants block above is replaced by:

259 Physical constants: HIDDEN. The numerical values of gravity, rod lengths, point
260 masses, and damping are NOT disclosed. You must infer dynamics from the initial
261 conditions alone.

262 **Parsing and scoring.** Each response is stripped of code fences and (for CoT) of everything outside the
263 `<answer>...</answer>` tags; the first $\{...\}$ object is JSON-parsed and validated to contain length- k theta
264 and omega arrays. A response that fails any check is counted as unanswered and scored at $\pi/2$.

265 B Training and compute

266 All primary learned models use a 256-wide, 3-layer tanh MLP, Adam, and a cosine-annealed learning rate, trained
267 on a single 8 GB consumer GPU. Mixed- k rollout adds negligible parameters; its cost is wall-clock (the HNN
268 mixed- k run is the most expensive at ~ 6.7 h because each unroll step recomputes ∇H via autograd). Chronos
269 runs at inference only. The dataset has 67,491 samples across $k \in \{1, 2, 3\}$ and three regimes, subsampled at
270 $\Delta t = 0.01$ s from the 10^{-3} s ground truth. Despite the number of samples, the training set contains only *nine*
271 trajectories, one for each (k, regime) cell. The budget study uses nested subsets of 1, 5, and 20 trajectories per
272 eligible cell and three training initializations. Neural ODE therefore sees 9, 45, or 180 trajectories; HNN sees 3,
273 15, or 60 undamped trajectories. Evaluation uses the same two held-out seeds throughout.

274 **Training objectives.** Derivative matching minimizes MSE on finite-difference $\dot{s}$; rollout loss minimizes
275 MSE on the state after k Euler steps with gradients through the unroll; mixed- k rollout trains on several window

lengths jointly (Neural ODE: $k \in \{10, 50\}$; HNN: $k \in \{5, 20\}$). The LNN’s loss requires second-order autodifferentiation and is trained at full scale (67k samples, hidden 256) on a CUDA GPU. Chronos-2 resamples the context to up to 512 points and forecasts ≤ 64 steps.

Reproducibility records. The anonymized artifact accompanying this submission contains the code, prompts, trained weights, held-out specifications, and the per-cell records for the main held-out comparison (2,160 records), the matched disclosure test (1,800), the modality and prompting grid, and the Lorenz sweep; the two large sweeps, the training-budget and balanced-seed studies, are packaged as separate archives. All dataset and held-out seeds are fixed and recorded in the released configuration. LLMs are sampled once at temperature 0. The reported hosted-model numbers recompute from the saved records; re-calling the API is a new evaluation rather than a reproduction. Invalid cells score $\pi/2$.

C Seed and system replication

Lorenz uses $(\sigma, \rho, \beta) = (10, 28, 8/3)$, RK4 ground truth at $\Delta t = 0.01$ s, a 5 s context, one fixed training seed, and five fixed held-out seeds (all recorded in the released configuration). Each held-out seed contains five trajectories, giving 25 held-out trajectories in total. The Neural ODE is a two-hidden-layer, width-128 tanh network trained for 300 epochs by analytic derivative matching. SINDy uses a degree-two polynomial library with STLSQ, fit to analytic derivatives at the sampled training states; the refit selects the true sparsity pattern exactly and recovers $(\hat{\sigma}, \hat{\rho}, \hat{\beta}) = (10, 28, 8/3)$ with maximum absolute coefficient error 2.2×10^{-13} . NVAR uses two delays, degree-two features, and ridge regression. Chronos-2 uses `amazon/chronos-2` in univariate and multivariate modes.

Table 4: Replication results. Top: reliability-adjusted pendulum error (rad) by held-out seed and equal-seed mean; horizons $\{1, 10\}$ s. The original seed has 20 trajectories per (k , regime) cell and the new seed has five. Equal-seed answer rates are 97% for LNN, 61% for SINDy, and 46% for NVAR; all other rows answer 100%. Bottom: Lorenz–63 normalized MAE, averaged within each of five held-out seeds (five trajectories each) and then across seeds, with the standard error over seeds. SINDy is below 2×10^{-5} in every cell. Lower is better.

Pendulum model	primary seed	second seed	seed mean
rk4 (reference)	0.059	0.074	0.067
symplectic	0.205	0.194	0.200
euler	0.405	0.415	0.410
chronos2-uni	0.697	0.604	0.650
chronos2-multi	0.684	0.621	0.653
sindy	0.860	0.906	0.883
hnn-rollout	0.972	0.858	0.915
neural-ode-rollout-mixed	1.036	0.989	1.012
hnn-rollout-mixed	1.066	1.037	1.051
nvar	1.245	1.251	1.248
lnn	1.356	1.376	1.366
Lorenz model	0.1 s	0.5 s	1.0 s
rk4 (reference)	0.000	0.000	0.000
sindy	$<10^{-5}$	$<10^{-5}$	$<2 \times 10^{-5}$
neural-ode	0.014 (0.008)	0.079 (0.030)	0.277 (0.063)
nvar	0.012 (0.001)	0.262 (0.050)	0.429 (0.084)
chronos2-uni	0.126 (0.027)	0.700 (0.102)	0.732 (0.085)
chronos2-multi	0.101 (0.018)	0.691 (0.129)	0.771 (0.097)
euler	0.028 (0.005)	0.216 (0.023)	0.885 (0.120)
persistence	0.524 (0.054)	1.028 (0.123)	0.900 (0.057)

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D Related work

The audited models span learned dynamics [Chen et al., 2018, Greydanus et al., 2019, Cranmer et al., 2020] on chaotic testbeds [Gilpin, 2021], zero-shot LLM and foundation-model forecasting [Gruver et al., 2023, Zhang and Gilpin, 2025, Ansari et al., 2024, Das et al., 2024, Woo et al., 2024], LLM physical reasoning [Lewkowycz et al., 2022, Wei et al., 2022, Xu et al., 2025], and sparse identification [Brunton et al., 2016, Gauthier et al., 2021]. The identifiability check draws on structural identifiability under parameter symmetries [Borgqvist et al.,

2024]. Unlike selective prediction, which reports risk together with coverage [Geifman and El-Yaniv, 2017], failed cells here stay in the main score because models are compared at fixed task coverage.

E Extended diagnostics

This appendix expands the diagnostics summarized in §4.5.

Behavior by horizon. Out-of-sample error by horizon appears in Table 12. At 1 s the ordering follows the main comparison: the integrators are near-exact, `kimi` (0.457) has the lowest black-box error, and the learned models are at 0.86–1.25. At 10 s all models are in the chaotic regime, and the ordering is stable: `kimi` (0.619) remains lowest among black-box methods while the learned models reach 1.09–1.46. The mixed- k Neural ODE is mid-pack at both horizons (0.951 / 1.120). Figure 1 shows the growth curves over the $\{0.01, 1, 10, 60\}$ s ladder: the numerical integrators stay near zero until ~ 10 s, whereas every black-box family saturates near the chaotic noise floor by 1 s. The $\{1, 10\}$ s window used for the comparison is the range over which the families remain separable.

Breakdown by k and regime. Tables 5 and 6 report out-of-sample error by number of links and by regime. Error increases monotonically with k for every family (for `kimi`, $0.11 \rightarrow 0.55 \rightarrow 0.95$; the integrators are near-exact at $k=1$ and degrade as chaos sets in), and the cross-family ordering holds within each k . The `changed_hidden` regime has the lowest error for almost every model (for `kimi`, 0.46 hidden vs. 0.64 normal). For the learned models, withholding the hidden constants reduces but does not remove the same difficulty effect: hidden-regime errors are now 0.71–1.19; the previous values favored models given the true constants. The low-gravity hidden constants ($g=1.62$) produce slower, less-divergent motion over a fixed horizon, so the unadjusted lower hidden-regime error contrast is a difficulty confound (§4.2). A cluster bootstrap over complete held-out trajectories changes no confidence interval by more than ~ 0.03 rad and leaves the ranking unchanged.

Training-objective check. Table 10 reports the effect of the loss with architecture fixed. Under the equalized hidden-constant analysis, rollout loss is not uniformly better: mixed- k rollout is lower for the Neural ODE, while single- k rollout is lower for the HNN. Full-scale CUDA training of the LNN gives the highest learned-model error (1.356), with 6% of cells diverging or exceeding the rollout timeout after memory-safe recovery. We make no architecture-wide claim about LNNs; under our implementation and protocol the LNN was numerically unstable and generalized poorly out of sample.

Physical-fidelity check. Hamiltonian and Lagrangian networks target structure preservation, so we evaluate energy drift on the undamped (`normal`) regime, where true energy is constant. Table 11 reports mean absolute drift over 10 s, normalized by $E^* = g \sum_i m_i L_i$, and the fraction of bounded rollouts. The learned models do not match classical integrators on this axis: the lowest-drift learned model is 0.78, compared with 0.066 for the symplectic integrator and 0.494 for Euler. Symplectic rollout of the HNN leaves drift nearly unchanged, so changing the integrator does not account for the HNN’s drift. At the derivative-matching objective, HNN drift is comparable to Neural ODE drift (4.27 vs 4.35); the LNN is unstable, with 27% bounded rollouts. These results use the same nine-trajectory learned-model budget as the main comparison.

Multivariate time-series check. On 720 paired samples, Chronos-2 multivariate and univariate errors are indistinguishable: 0.543 (95% CI [0.498, 0.589]) versus 0.549 ([0.505, 0.593]). No effect remains after Holm correction, so the larger study rejects the apparent per- k trend in the primary grid.

Predictability horizon. Pointwise error past the Lyapunov time saturates to a system-dependent constant for every model, so long-horizon error rankings carry little information. We define the predictability horizon as the longest horizon a model tracks contiguously below a threshold τ from $t=0$; the contiguous rule is used because, after divergence, a later horizon can fall below τ by angle-wrap coincidence. Table 13 sweeps τ .

F Breakdowns by link count and regime

Reliability-adjusted mean angle error (rad) on the shared out-of-sample held-out grid (horizons $\{1, 10\}$ s), by number of links (Table 5) and by regime (Table 6); models in leaderboard order. Cells a model never produced (e.g. the LNN’s diverged $k=3$) are scored at $\pi/2$. Each row uses the same grid as Table 2.

G Additional result tables

Full per-model values for the horizon breakdown and the predictability horizon (both defined in Appendix E), together with the regime definitions, the matched disclosure gap, and the training-budget, training-objective, and physical-fidelity studies. The corresponding summary statistics appear in the main text.

Table 5: Out-of-sample error by number of links k .

Model	$k=1$	$k=2$	$k=3$
rk4 (reference)	0.000	0.068	0.109
symplectic	0.000	0.253	0.363
euler	0.271	0.410	0.534
kimi-k2.6	0.112	0.553	0.949
chronos2-multi	0.263	0.798	0.992
chronos2-uni	0.279	0.813	0.999
grok-4-1-fast-reasoning	0.366	1.030	1.247
deepseek-v4-pro	0.872	0.949	1.036
hnn-rollout	0.524	0.985	1.406
neural-ode-rollout-mixed	0.845	1.178	1.084
hnn-rollout-mixed	0.550	1.162	1.487
neural-ode	0.626	1.304	1.400
neural-ode-rollout	0.968	1.172	1.224
hnn	0.511	1.337	1.567
lnn	1.072	1.458	1.517

Table 6: Out-of-sample error by regime. Learned-model hidden cells use the equalized no-hidden-constants run. The hidden regime remains easier for several models because its low-gravity constants give slower, less-divergent motion. The unadjusted regime contrast in §4.2 consequently mixes disclosure with difficulty.

Model	normal	disclosed	hidden
rk4 (reference)	0.174	0.003	0.000
symplectic	0.387	0.177	0.051
euler	0.743	0.387	0.085
kimi-k2.6	0.635	0.519	0.461
chronos2-multi	0.818	0.826	0.409
chronos2-uni	0.818	0.798	0.475
grok-4-1-fast-reasoning	1.096	0.959	0.588
deepseek-v4-pro	1.240	1.097	0.520
hnn-rollout	1.087	1.115	0.713
neural-ode-rollout-mixed	1.260	0.836	1.011
hnn-rollout-mixed	1.244	1.172	0.782
neural-ode	1.221	1.095	1.014
neural-ode-rollout	1.281	0.894	1.189
hnn	1.262	1.303	0.850
lnn	1.511	1.408	1.127

Table 7: Matched disclosure gap (rad), paired on identical trajectories. $\Delta_{\text{disc}} = \text{err}(\text{hidden}) - \text{err}(\text{disclosed})$; positive values indicate higher error without the constants. Reliability-adjusted; paired Wilcoxon, Holm-corrected. This experiment uses its own matched grid (3 models $\times$ 2 horizons $\times$ 150 trajectories = 900 pairs), not the shared held-out set, so its absolute errors are not comparable to Table 6.

Model	disclosed	hidden	Δ_{disc}
kimi-k2.6	0.746	0.957	+0.211
grok-4-1-fast-reasoning	0.987	1.025	+0.038
deepseek-v4-pro	1.425	1.382	−0.042
pooled (paired, $n=900$)			+0.069*
$k=1$			+0.297***
$k=3$			−0.055*

* $p_{\text{Holm}} < 0.05$, *** $p_{\text{Holm}} < 10^{-3}$.

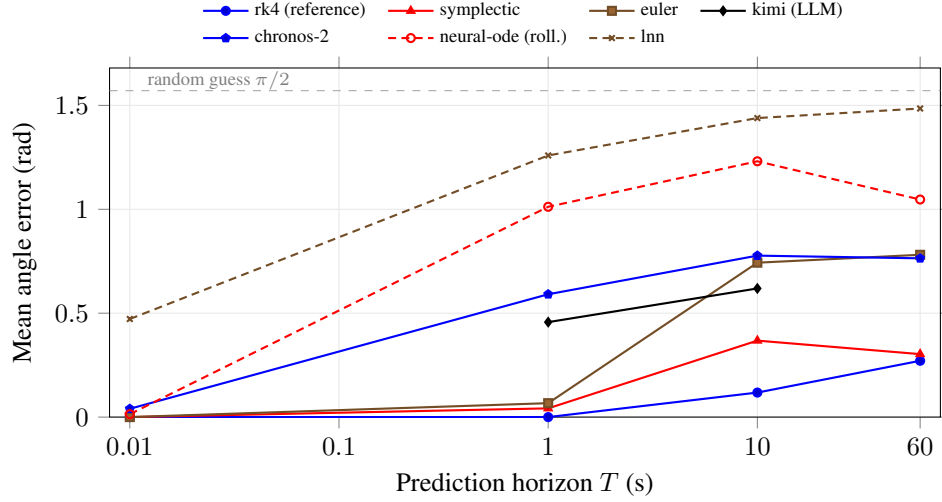


Figure 1: Error growth with prediction horizon (out-of-sample, reliability-adjusted; log time axis). The numerical integrators stay near zero until ~ 10 s; every black-box family saturates near the chaotic noise floor by 1 s. LLMs were evaluated at $\{1, 10\}$ s only. Dashed line: random-guess baseline $\pi/2$.

Table 8: Physical regimes. `changed_hidden` withholds the constants.

Regime	g (m/s ²)	L (m)	m (kg)	damping	disclosed
normal	9.81	1.0	1.0	0.0	—
changed_disclosed	3.71	1.4/0.7/1.1	2.0/0.6/1.5	0.05	yes
changed_hidden	1.62	0.8/1.2/0.9	0.5/1.8/1.0	0.10	no

Table 9: Nested training-budget study. Reliability-adjusted pendulum error (rad), averaged over three training initializations and two held-out seeds. Intervals hierarchically resample initializations and held-out seeds. HNN is evaluated on the undamped normal regime only.

Model	Scope	Trajectories	Error	95% CI
HNN	normal	3	1.269	[1.209, 1.349]
HNN	normal	15	0.974	[0.891, 1.062]
HNN	normal	60	0.981	[0.856, 1.113]
Neural ODE	all regimes	9	1.090	[1.011, 1.207]
Neural ODE	all regimes	45	0.968	[0.923, 1.028]
Neural ODE	all regimes	180	0.861	[0.829, 0.907]

Table 10: Training-objective study, out-of-sample mean angle error (rad) on the shared held-out grid (horizons $\{1, 10\}$ s); architecture fixed, loss/window varied. Significance from a powered paired study on the full four-horizon held-out set ($n=720$, Wilcoxon+Holm).

Model	Objective	Err	vs. prev. row
Neural ODE	derivative MSE	1.110	—
	rollout (single- k)	1.121	n.s. ($p_{\text{Holm}}=0.188$)
	mixed- k	1.036	sig. ($p_{\text{Holm}}=0.013$)
HNN	derivative MSE	1.138	—
	rollout (single- k)	0.972	sig. ($p_{\text{Holm}}=1.95 \times 10^{-4}$)
	mixed- k	1.066	sig. higher ($p_{\text{Holm}}=0.014$)

Table 11: Physical fidelity on the undamped (normal) regime: mean absolute energy drift over a 10 s rollout, normalized by $E^* = g \sum_i m_i L_i$ (0 = perfect conservation), and fraction of bounded rollouts. HNN drift is nearly unchanged when its rollout uses a symplectic integrator.

Model	Family	Energy drift	Stable
rk4	numerical	0.000	100%
symplectic	numerical	0.066	100%
euler	numerical	0.494	100%
neural-ode-rollout-mixed	learned	0.779	100%
hnn-rollout-mixed	learned	0.865	100%
neural-ode-rollout	learned	0.872	100%
hnn-rollout	learned	2.065	100%
lnn	learned	4.038	27%
hnn	learned	4.270	100%
neural-ode	learned	4.346	100%

Table 12: Out-of-sample mean angle error (rad) by horizon, reliability-adjusted, on the shared held-out grid (selected models). The cross-family comparison uses horizons $\{1, 10\}$ s.

Model	1 s	10 s
rk4 (reference)	0.000	0.118
symplectic	0.042	0.368
euler	0.067	0.743
kimi-k2.6	0.457	0.619
chronos2-multi	0.591	0.777
chronos2-uni	0.622	0.773
grok-4-1-fast-reasoning	0.905	0.857
deepseek-v4-pro	0.996	0.909
hnn-rollout	0.855	1.088
neural-ode-rollout-mixed	0.951	1.120
hnn-rollout-mixed	0.982	1.150
neural-ode	0.892	1.328
neural-ode-rollout	1.012	1.231
hnn	1.052	1.224
lnn (full-scale)	1.248	1.464

Table 13: Predictability horizon (s): longest horizon tracked contiguously below threshold τ , on the shared out-of-sample set (ladder $\{0.01, 1, 10, 60\}$ s). Higher is better; “—” means no horizon, including the first, stayed below τ .

Model	$\tau=0.1$	0.25	0.5	1.0
rk4 (reference)	1.0	10	60	60
symplectic	1.0	1.0	60	60
euler	1.0	1.0	1.0	60
neural-ode-rollout	0.01	0.01	0.01	0.01
neural-ode-rollout-mixed	0.01	0.01	0.01	1.0
chronos2-multi	0.01	0.01	0.01	60
chronos2-uni	0.01	0.01	0.01	60
hnn-rollout	0.01	0.01	0.01	1.0
neural-ode	0.01	0.01	0.01	1.0
hnn	0.01	0.01	0.01	0.01
lnn	—	—	0.01	0.01